
A Validated Black-Box Instrument for Language-Model Dynamics: Reproducing Known Metrics with a Token-Lattice Cellular Automaton

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Abstract

1 We turn a language model into a *cellular automaton over token space* — a ring
2 of token cells resampled in place by the model’s own windowed conditional
3 $p_r(x_i \mid x_{i \pm r})$, probed by common-random-number damage spreading — and de-
4 velop it into a *validated* measurement instrument, on the principle of **validation**
5 **by reproduction**. On the Domany–Kinzel automaton, where the damage field is
6 provably the automaton itself, an independent prediction matches ours **bit-exactly**
7 — zero mismatching cells, with a nonzero off-line control — a calibration with
8 no error bar attached, which a plausible-looking but wrong implementation cannot
9 pass. The instrument also separates elementary CA rules by whether damage sur-
10 vives ($d=3.0$) and recovers known Markov matrices. Across Pythia-410m check-
11 points it detects a **developmental transition**: the token-space Lyapunov exponent
12 goes from seeds disagreeing about its *sign* to **every one of 48 post-transition**
13 **runs being positive**, at two lattice sizes, all four members of a pre-registered fam-
14 ily surviving Benjamini–Hochberg correction. We then *delimit* it, the result we
15 consider most useful: real autoregressive generation absorbs *no* injected token er-
16 ror ($P_{\text{persist}}=1.000$ on three models), because free generation never resamples a
17 token. These quantities characterise an iterated-resampling construction, not de-
18 ployed decoding — a boundary we draw rather than leave to the reader.

19 1 Introduction

20 Language models are evaluated almost entirely as functions: given a context, how good is the next-
21 token distribution? But a masked or autoregressive conditional also *defines a dynamics* — iterate it
22 in place over a lattice of tokens and one obtains a stochastic cellular automaton whose rule is the
23 model. The phenomena a CA vocabulary names — damage spreading [Bagnoli et al., 1992], light
24 cones [Lieb and Robinson, 1972], Lyapunov exponents, attractor censuses [Hanson and Crutchfield,
25 1997] — are decades old. What has been missing is not a phenomenon but a *measurement instrument*
26 *whose readings can be trusted*.

27 Our organizing principle is therefore **validation by reproduction** (§3): before turning the appara-
28 tus on a language model we require it to recover values that are independently known. Damage-
29 spreading implementations are conventionally validated against fitted critical exponents, which a
30 wrong implementation can match for the wrong reasons; our strongest rung is instead *exact*, and a
31 bit-exact identity cannot be passed by a plausible-looking estimator. Only then do we report what
32 the instrument measures on trained LMs (§4) and, equally, where its readings provably stop applying
33 (§5, §6).

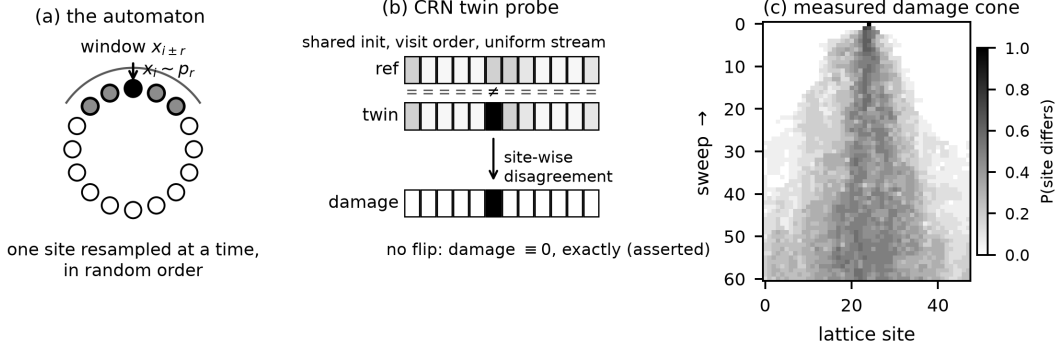


Figure 1: The instrument. (a) N token cells on a ring, async random-order single-site Glauber; the windowed conditional resamples one cell, special tokens forbidden. (b) CRN probe: twins share initial state, visit order and uniform stream, so the *null* arm must differ in exactly zero cells — asserted on every backend — and the probe arm’s site-wise disagreement after one flipped token is the damage field. (c) A measured damage cone ($T=0.7$, $r=1$): site-wise disagreement probability spreading from the single flipped site.

2 The instrument

Rule (Fig. 1). A window of $w=2r+1$ tokens with the centre masked is fed to a masked LM; the tempered centre distribution is the kernel $p_r(x_i | x_{i \pm r})$. For an autoregressive model the window is the r cells to the left. **Confounders.** A degenerate low-entropy lattice snaps a perturbation back trivially, so damage D is normalised by the model’s own diversity floor D_0 (twins under *independent* noise, no flip), giving $D_{\text{norm}}=D/D_0$. The floor’s coupling cost is bounded (App. B). A free control disciplines our language: the temperature “phase transition” is a finite-size *crossover* ($\chi_{\text{peak}} \propto 1/N$).

3 Validation: reproducing known metrics

We check the apparatus against systems where the answer is known, ordered by how much of the instrument’s regime each rung shares (Fig. 3, App. B). (1–2) are smooth-limit unit tests, labelled as such: the logistic map’s analytic exponent is recovered to machine precision and a coupled-map lattice matches an exact Benettin reference to 1.1×10^{-3} ; neither is evidence about damage spreading — the twin is re-anchored each step, and a **token flip** is $O(1)$.

(3) **Elementary CA: a coarse split.** The identical protocol on 19 rules of *known* class (12 seeds, rule as the unit) separates damage that *survives* from damage that *dies*. The discriminating quantity is the **ignition probability** — also the directed-percolation order parameter [Grassberger, 1995] — not λ , which averages a bimodal mixture. Ordered rules sit at 0.046 (CI [0.000, 0.102]) against 0.668 (edge) and 0.682 (chaotic): **ordered vs. rest** $p=0.0$, $d=3.03$. The three-class ordering does *not* survive ($p=0.470$), so we claim only the coarse split; the demoted group-mean λ ordering, and why it was never a measurement, are in App. B.

(4) **Domany–Kinzel: an exact check.** DK [Domany and Kinzel, 1984] is the only rung that is stochastic *and* discrete — the instrument’s own regime — and on its $p_2=0$ line the damage field between two commonly-driven replicas is provably *itself* a DK automaton at the same p_1 [Kohring and Schreckenberg, 1992]. We therefore predict the damage trajectory with an independent simulator and demand *bit-identity*: across $p_1 \in \{0.2, \dots, 1.0\}$ on a ring of 4096 over 1500 steps, **zero mismatching cells**, against 16 in an off-line control at (0.6, 0.5) where the mapping must break. It runs through the same loop as every language-model number below, so indexing, uniform consumption, sampling and schedule are verified with no error bar attached; Rule 90 rides inside the same run as a second closed form (0.03125 live cells at $t=1500$, as obtained). Critical points are a weaker $\sim 1\%$ check: $p_c=0.7065$ on the site-DP line (published 0.705489(4)) and 0.8092 on $p_2=0$, the damage field independently at 0.8089. The published $p_2=0$ value is disputed (0.801(2) [Zebende and Penna, 1994] vs. 0.8087(5) [Hinrichsen et al., 1997]), so we report both and claim neither.

Table 1: The pre-registered developmental family: means over the pre-registered sets; d is Cohen’s d ; λ_{ca} rows exclude unignited runs ($n_{pre}=15$ at $N=96$); $<10^{-6}$ means the stored p rounds to zero at six decimals.

quantity	N	pre mean	post mean	d	p_{BH}
λ_{ca}	48	+0.0247	+0.1683	1.59	1.7×10^{-5}
λ_{ca}	96	+0.0320	+0.1686	1.71	2.3×10^{-5}
D_{norm}	48	0.1865	0.5689	2.88	$< 10^{-6}$
D_{norm}	96	0.1030	0.3062	3.07	$< 10^{-6}$

(5) **Synthetic Markov sources.** The attractor census recovers each model’s own sparse transition matrix (row total-variation 0.22) and not the others (cross 0.95; baseline 0.91).

4 A developmental transition in token-space criticality

Pointed at a training run, the calibrated instrument finds a developmental transition: token-space dynamics cross from sub- to super-critical early in training and do not return. Across Pythia-410m checkpoints (step256–step143000, 8 seeds, $N=48$ and 96; Fig. 2) the crossing lies between steps 256 and 512 at both lattice sizes, λ_{ca} staying positive thereafter.

The effect occupies a temperature window, and the window has a mechanism. Repeating the two ends of the curve at $T \in \{0.3, 0.5, 0.9, 1.1\}$ (8 seeds each, BH-FDR over the four) recovers the transition at $T=0.5$ ($p_{BH}=6 \times 10^{-4}$) alongside the $T=0.7$ of the main grid, and at none of 0.3, 0.9, 1.1 ($p_{BH}=0.59, 0.72, 0.59$). The ignition fraction makes that a floor and a ceiling rather than a failure: below the window damage barely propagates at either end and λ_{ca} hugs zero, whereas above it the lattice is *already* super-critical before the training being measured, leaving nothing to move; inside the window it swings $0.23 \rightarrow 0.80$ at $T=0.5$ (floor and ceiling values in App. B). Independent work puts Pythia *generation*’s critical temperature at $T_c \approx 1$ [Nakaishi et al., 2024]; a different sampler, so a consistent reference point rather than the same measurement — but a ceiling between 0.7 and 0.9 is where it predicts one.

The test was pre-registered (post- vs pre-transition, run-level Mann–Whitney, family = $\{\lambda_{ca}, D_{norm}\} \times \{N=48, 96\}$, BH-FDR) with a demotion rule: failure at either size demotes the headline. **All four survive** ($p_{BH} \leq 2 \times 10^{-5}$; Table 1). The pre set is 256/512 and the plateau 2000/8000/143000; one $N=96$ run is excluded because its damage never ignited, so λ_{ca} is undefined for it.

The claim is better stated without any pre/post split, since every effect size depends on where that line is drawn. Before the transition the seeds do not agree on the *sign* of λ_{ca} (6/16 negative at $N=48$, 7/15 at $N=96$, spanning -0.216 to $+0.320$); after it, **not one of 48 plateau runs is negative**, the minimum being $+0.107$. That is ordinal and uses every run (robustness history in App. B).

The curve is not a step, and both ends can be inflated. It overshoots at step 1000; taking the peak as the post value, or the lowest checkpoint alone as the pre value, inflates the effect, so both ends use the pre-registered sets. The overshoot survives correction in one cell of four (accounting in App. B), so step 1000 is no distinct phase.

It is not specific to the model we picked. Repeating the grid on four Pythia sizes (70m/160m/410m/1b $\times$ 6 checkpoints $\times$ 8 seeds, 192 runs) reproduces the transition in **all four** (per-size p_{BH} in App. B). Its *timing* moves later with size and then saturates: 70m is already super-critical at the earliest checkpoint probed, 160m crosses between steps 128 and 256, and 410m and 1b both cross between 256 and 512. Learning rate is confounded with size across the suite (rates in App. B) and the two sizes sharing a rate are the two sharing a bracket, so we report the shift as an ordering, not a size effect. An orthogonal instrument agrees on the onset: Nakaishi et al. [2024] place the emergence of critical structure in Pythia-160m at $k_c \approx 10^2$ steps by power spectra and part-of-speech correlation — no damage spreading — adjacent to our 128–256 bracket. Both grids are coarse there, so this is agreement to within about a factor of two, not a reproduction. The plateau *level* does not order with size (levels in App. B); at four sizes the smallest attainable permutation p is $1/4! = 0.042$, so no capacity axis could have been established, and none is visible. We report that null as the closest thing here to a previously retracted capacity claim.

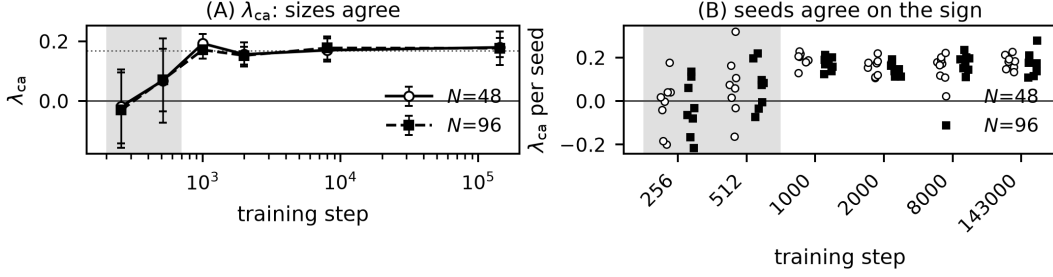


Figure 2: The developmental transition; shading marks the pre-registered pre set. (A) λ_{ca} rises and settles positive, the two sizes on top of each other (levels agree within $\pm 14\%$). (B) Per-seed: sign disagreement before, consensus after; D_{norm} matches at a size-dependent level.

110 **It is not a restatement of the loss curve.** The transition sits where everything else in training
 111 changes, so λ_{ca} might be an expensive perplexity proxy. It is not: held-out loss falls *monotonically*
 112 at all four sizes while λ_{ca} overshoots at three, and a non-monotone function of a monotone variable
 113 is not a monotone transform of it. They also disagree about *where* (details in App. B). We use the
 114 curve’s *shape* and not its level, the evaluation text not being Pythia’s training data.

115 **Which quantity carries the claim.** A third lattice size settles it. Over $N=48/96/192$ the λ_{ca}
 116 plateau is **intensive across a $4\times$ range** — slope $N^{-0.04}$, varying by 5% of its mean — while
 117 D_{norm} falls as $N^{-1.02}$, i.e. $1/N$ (levels in App. B). That is what the construction predicts: λ_{ca} is a
 118 cone-growth *rate* fitted before saturation; D_{norm} is a density ratio whose numerator stays in a cone
 119 while its denominator is delocalised. So λ_{ca} carries the claim and D_{norm} corroborates at a stated
 120 lattice size, never as a lattice-free property. Seed-to-seed variance also collapses $\sim 3\times$ across the
 121 transition (Levene $p \leq 1.7 \times 10^{-4}$ at both sizes), an observation rather than a pre-registered test.
 122 λ_{ca} is continuous and unthresholded, so this sharpening is not an artefact of a discontinuous metric
 123 [Schaeffer et al., 2023].

124 5 What the damping length measures: free generation absorbs nothing

125 We tested the instrument’s central assumption — that a ring number says something about the model
 126 — on *real autoregressive generation*: two CRN twins from a prompt sharing one uniform stream,
 127 one forced to a different token at one position, both continuing freely. The null arm diverges by
 128 exactly zero, as asserted.

129 The result is unambiguous and negative. On pythia-70m, -160m and -410m (32 trials each) the
 130 injected error **persists in every single trial** ($P_{persist}=1.000$). Token identity is harsh once an injection
 131 shifts alignment, so we also measured distributionally: twin next-token TV, normalised by that
 132 between *independent* continuations, settles at $TV_{norm} \approx 0.97$.

133 The mechanism is structural. **Free generation never resamples a token:** once emitted an error
 134 stays in context permanently, whereas the ring CA revisits every site, and in-place resampling is
 135 what makes healing possible. The damping length, D_{norm} and the light cone therefore characterise
 136 *the iterated-resampling construction* and not decoding — which also explains a light-cone velocity
 137 depending on r but not the model, and the failure in §6. It does *not* touch §4: the construction
 138 is held *fixed* across checkpoints — same ring, radius, temperature, coupling, seeds — so a change
 139 measured across checkpoints cannot come from the apparatus. The licence is therefore *comparative*,
 140 checkpoint against checkpoint, and never absolute: we do not claim any model *is* critical, since that
 141 would need the ring, the radius and D_0 to be principled rather than merely fixed, and they are
 142 choices.

143 Damage spreading is a property of (*model, coupling*), not of the model alone [Kohring and Schreck-
 144 enberg, 1992, Grassberger, 1995]; our coupling choice, its justification and its measured cost are
 145 stated precisely in App. B, and relative comparisons are unaffected because the coupling is a com-
 146 mon mode.

6 The instrument’s boundary: token- vs. activation-space

A black-box instrument earns its keep if its readings track what internals would otherwise be needed to see. Across two model families and two within-model axes they do not: the pairing is family-dependent (Pythia $r=+0.71$, GPT-2 $r=-0.43$; the pooled $p=0.025$ is a Simpson’s-paradox artifact), the temperature axis is confounded, and the radius axis is null because $\lambda_{ca}(r)$ is model-invariant. The reason is structural: the white-box exponent is architectural and *tangent*, while λ_{ca} is *finite*. A clean **negative** — the two levels are distinct and non-proxying.

7 Related work

Iterated-LLM views exist qualitatively [Utimula, 2026, Perez et al., 2024, Wang et al., 2025] and rigorously for masked-LM Glauber dynamics [Sana et al., 2026]; our novelty is the measurement layer, not the framing. Edge-of-chaos measurement is canonical [Langton, 1990, Bertschinger and Natschlager, 2004] and already applied to transformers [Tomihari and Karakida, 2025] and LLMs [Zhang et al., 2025, Li et al., 2025]; we claim neither the concept nor the criticality $\leftrightarrow$ capability link, and absorbing-state scaling laws in deep networks are established [Tamai et al., 2025] — we report no universality class. Temperature-driven order–disorder transitions have several prior treatments [Ruan et al., 2026, Sun and Haghighat, 2025, George et al., 2025, Nakaishi et al., 2024, Arnold et al., 2024]; we claim a calibrated route, not the quantity. Perturbation propagation in *activation* space [Petrova and Vejsiu, 2026, Olivieri and Perez Rodrıguez, 2026, Fernando and Guitchounts, 2026] and over compound-AI graphs [Nilayam and Nayak, 2026] is adjacent but in a different space; we avoid “repair” [McGrath et al., 2023, Rushing and Nanda, 2024] and “self-correction” [Petrova and Vejsiu, 2026], categorically different from a spatial damping length. Tang et al. [2026] is the calibration kin; the census descends from Hanson and Crutchfield [1997].

8 Limitations

Neither construction samples the model’s actual distribution — masked conditionals are globally inconsistent, the autoregressive rule is resampled in place on a ring — so all claims are properties of the dynamics. Rings are small ($N \leq 384$); the developmental result rests on one model *family* (four sizes and three lattice sizes within it, so a second family is the next requirement) and on a narrow temperature window, outside which the probe saturates either way (§4). D_{norm} ’s absolute scale is doubly qualified — it moves with N , and its numerator’s coupling is not extremal — so only its relative behaviour is quoted, and census ground truth is synthetic only. **Statistical scope.** The developmental family is pre-registered with BH-FDR at the run level and the ECA test uses the rule as its unit; λ_{ca} statistics exclude runs whose damage never ignited, for which it is undefined, with n stated. The ladder rests on no hypothesis test and the cross-level negative on sign-consistency rather than a surviving p -value. A previously reported capacity $\rightarrow$ sensitivity axis was pseudoreplicated and is retracted.

9 Responsible use and conclusion

This instrument measures a construction built *from* a language model, from sampled outputs alone. That is its affordance: it characterises models behind an API, where weights are unavailable and activation-space methods cannot reach. It creates no new generative capability. The misuse we consider most likely is a misreading of our own numbers, which is why §5 is in the body: a damping length measured here describes an in-place-resampling ring, and real generation absorbs *no* injected token error. These quantities must **not** be cited as evidence about the robustness, error tolerance or self-correction of a deployed decoder — a λ_{ca} below zero is not a safety property. That boundary generalises: a metric should be made to reproduce known answers before it is believed about unknown ones. We offer the ladder as a method, not only as our step one — it demoted our own ECA ordering and caught a λ_{ca} reported where it is undefined.

References

- Julian Arnold, Flemming Holtorf, Frank Schäfer, and Niels Lörch. Phase transitions in the output distribution of large language models, 2024.
- Franco Bagnoli, Raúl Rechtman, and Stefano Ruffo. Damage spreading and lyapunov exponents in cellular automata. *Physics Letters A*, 172(1–2):34–38, 1992.
- Nils Bertschinger and Thomas Natschläger. Real-time computation at the edge of chaos in recurrent neural networks. *Neural Computation*, 16(7):1413–1436, 2004.
- Stella Biderman, Hailey Schoelkopf, Quentin Anthony, Herbie Bradley, et al. Pythia: A suite for analyzing large language models across training and scaling, 2023.
- Eytan Domany and Wolfgang Kinzel. Equivalence of cellular automata to Ising models and directed percolation. *Physical Review Letters*, 53(4):311–314, 1984.
- Jesseba Fernando and Grigori Gutichounts. Dynamics of the transformer residual stream: Coupling spectral geometry to network topology, 2026.
- Jackson George, Zachariah Yusaf, Stephanie Zoltick, and Linh Huynh. (un)biased data and spin glasses reveal clustering for Turing phase transitions within human–transformer interactions, 2025.
- Peter Grassberger. Are damage spreading transitions generically in the universality class of directed percolation? *Journal of Statistical Physics*, 79(1):13–23, 1995.
- James E. Hanson and James P. Crutchfield. Computational mechanics of cellular automata: An example. *Physica D: Nonlinear Phenomena*, 103(1–4):169–189, 1997.
- Haye Hinrichsen, Joshua S. Weitz, and Eytan Domany. An algorithm-independent definition of damage spreading—application to directed percolation. *Journal of Statistical Physics*, 88(3): 617–636, 1997.
- G. A. Kohring and M. Schreckenberg. The Domany–Kinzel cellular automaton revisited. *Journal de Physique I*, 2(10):2033–2037, 1992.
- Christopher G. Langton. Computation at the edge of chaos: Phase transitions and emergent computation. *Physica D: Nonlinear Phenomena*, 42(1–3):12–37, 1990.
- Xiaojian Li et al. Cognitive activation and chaotic dynamics in large language models: A quasi-lyapunov analysis of reasoning mechanisms, 2025.
- Elliott H. Lieb and Derek W. Robinson. The finite group velocity of quantum spin systems. *Communications in Mathematical Physics*, 28(3):251–257, 1972.
- Thomas McGrath, Matthew Rahtz, Janos Kramar, Vladimir Mikulik, and Shane Legg. The hydra effect: Emergent self-repair in language model computations, 2023.
- Kai Nakaishi, Yoshihiko Nishikawa, and Koji Hukushima. Phase transition in large language models and the criticality of natural languages, 2024.
- Prashanti Nilayam and Sankalp Nayak. Quiver: A formal framework for quantifying perturbation propagation and bifurcation in compound ai systems, 2026.
- David N. Olivieri and Antonio F. Pérez Rodríguez. Transformer field theory: A response-theoretic approach to mechanistic interpretability, 2026.
- Jérémy Perez, Grgur Kovač, Corentin Léger, Cédric Colas, Gaia Molinaro, Maxime Derex, Pierre-Yves Oudeyer, and Clément Moulin-Frier. When LLMs play the telephone game: Cultural attractors as conceptual tools to evaluate LLMs in multi-turn settings, 2024.
- Ingrid Petrova and Luan Vejsiu. Spectral origins of the self-correction blind spot in autoregressive generation, 2026.

- 237 Huajian Ruan, Jinyang Li, Xingyu Guo, and Lingxiao Wang. Generative criticality in large language
238 model temperature scaling, 2026.
- 239 Cody Rushing and Neel Nanda. Explorations of self-repair in language models. In *International
240 Conference on Machine Learning (ICML)*, 2024.
- 241 Suvadip Sana, Sami Wolf, Neer Mehta, Alina Shah, Aitzaz Shaikh, Janna Goodman, and Lionel
242 Levine. Mixing times of glauber dynamics on masked language models, 2026.
- 243 Rylan Schaeffer, Brando Miranda, and Sanmi Koyejo. Are emergent abilities of large language
244 models a mirage?, 2023.
- 245 Youran Sun and Babak Haghighat. Phase transitions in large language models and the $o(n)$ model,
246 2025.
- 247 Keiichi Tamai, Tsuyoshi Okubo, Truong Vinh Truong Duy, Naotake Natori, and Synge Todo. Uni-
248 versal scaling laws of absorbing phase transitions in artificial deep neural networks. *Physical
249 Review Research*, 7:033072, 2025.
- 250 Luhan Tang, Longxuan Yu, Shaorong Zhang, and Greg Ver Steeg. Is your diffusion sampler actually
251 correct? a sampler-centric evaluation of discrete diffusion language models, 2026.
- 252 Akiyoshi Tomihari and Ryo Karakida. Recurrent self-attention dynamics: An energy-agnostic per-
253 spective from jacobians, 2025.
- 254 Keishu Utimula. Logos-ca: A cellular automaton using natural language as state and rule, 2026.
- 255 Zhilin Wang, Yafu Li, Jianhao Yan, Yu Cheng, and Yue Zhang. Unveiling attractor cycles in large
256 language models: A dynamical systems view of successive paraphrasing, 2025.
- 257 G. F. Zebende and T. J. P. Penna. The Domany–Kinzel cellular automaton phase diagram. *Journal
258 of Statistical Physics*, 74(5):1273–1279, 1994.
- 259 Shiyang Zhang, Aakash Patel, Syed A. Rizvi, Nianchen Liu, Sizhuang He, Amin Karbasi, Emanuele
260 Zappala, and David van Dijk. Intelligence at the edge of chaos. In *International Conference on
261 Learning Representations (ICLR)*, 2025.

262 A Reproducibility

263 **Models.** bert-tiny/mini/base ($L=2-24$); Pythia-14M-1B [Biderman et al., 2023] and GPT-2
264 small-xl; Pythia-410m checkpoints step256–step143000. A tiny transformer trained from
265 scratch on tinyshakespeare provides the corpus-checked census. **Grids.** $r \in \{1, 2, 4, 8, 16\}$,
266 $T \in \{0.3, \dots, 1.5\}$, $N \in \{48, 96, 192, 384\}$, 8 seeds for the developmental family. **Definitions.**
267 $D_{\text{norm}} = D/D_0$ with D_0 the independent-noise no-flip floor; λ_{ca} the log-slope of damaged-site
268 count over the pre-saturation window, with the fit window’s constants exposed as parameters and
269 a named sentinel for never-ignited runs. **Tests.** The exact-zero CRN null is asserted over every
270 backend and both update modes through one shared simulation loop; a golden-file regression as-
271 serts the core stays bit-identical; the DK identity is asserted cell-for-cell with an off-line control
272 that must fail. All code, per-run result files, and figure scripts accompany the submission; every
273 number in the paper is traceable to a result file. **Mirror.** An anonymised snapshot of the tagged
274 tree is at https://osf.io/9gpu2/?view_only=9973f67d79e7403e96cf4586d6efbdc0, built
275 from the submission tag by build_mirror.py, which rewrites absolute checkout paths and then
276 machine-checks the result for identifying strings.

277 B Robustness details

278 Numbers relocated from the body for density; every value below is unchanged from the submitted
279 version and traced to its results file by the number manifest.

280 **Four-size scale grid (§4):** the transition reproduces at every size, and the plateau level does not order
281 with size. Learning rates across the suite: 1.0×10^{-3} , 6.0×10^{-4} , 3.0×10^{-4} , 3.0×10^{-4} [Biderman
282 et al., 2023].

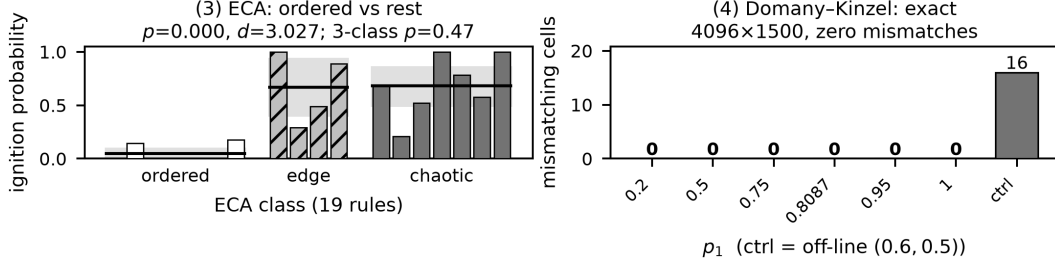


Figure 3: The two rungs that carry the calibration, numbered as in the text. (4) is the load bearing one: the DK damage field is predicted by an independent simulator and matches cell-for-cell, the off-line control beside it showing the test is not vacuous. Rungs (1–2) and (5) are reported in the text.

size	p_{BH}	plateau λ_{ca}
70m	0.015	0.162
160m	0.003	0.164
410m	2×10^{-6}	0.174
1b	1.5×10^{-5}	0.166

Three lattice sizes (§4): over $N=48/96/192$ the λ_{ca} plateau is 0.168/0.169/0.160 while D_{norm} is 0.569/0.306/0.139 — the levels behind the body’s $N^{-0.04}$ and $N^{-1.02}$ slopes. Seed-to-seed spread at the plateau: per-seed CV 22–25%; it is the checkpoint *means* that agree to a few percent across sizes.

Overshoot accounting (§4): the curve overshoots at step 1000 by 1.4–22.4% across the four cells; taking the lowest checkpoint alone as the pre value would inflate the λ_{ca} effect $1.7\times$, which is why both ends use the pre-registered sets. The overshoot survives BH correction only in the $N=48$ D_{norm} cell; on λ_{ca} it is $+1.4\%$ at $N=96$ ($p_{BH}=0.78$).

Loss baseline (§4): held-out loss at 26 (model, checkpoint) pairs — the scale grid’s 24 plus 410m at steps 8000 and 143000. The steepest loss descent is the 512–1000 bracket for *every* size, whereas the λ_{ca} crossing moves with size and precedes it. Rank correlation is mostly not significant at six checkpoints (ρ from -0.77 to -0.71), so shape and location carry this and not the correlation.

Temperature window (§4): the floor at $T=0.3$ — ignition $0.20 \rightarrow 0.21$ across training, λ_{ca} hugging zero — and the ceiling at $T=0.9$ and 1.1 , where pre-training λ_{ca} is already $+0.19$ and $+0.30$ with ignition 0.65 and 0.98.

Coupling, stated precisely (§2, §5): the diversity floor D_0 necessarily uses a different coupling from the damage numerator — a floor sharing the numerator’s draws is identically zero, since identical twins stay identical — and the cost is bounded: sweeping the shared-draw fraction from 0 to 0.9 moves D_{norm} by 4%. Our inverse-CDF sampling is the *monotone* coupling. On DK’s binary alphabet it coincides with the maximal-correlation member of the admissible family [Hinrichsen et al., 1997], which is why the DK identity is exact; on a vocabulary it does not, so the LM damage numbers sit inside the family rather than at an extremum — excess disagreement 1.3–5.4% across the temperature sweep. The choice is justified by replica-independence — each replica’s next state depends on its own state and the shared noise alone — which a pairwise maximal coupling lacks. One fact explains both ends of this paper: damage-spreading theory was developed on *binary* alphabets, which is why the DK rung is exact and why its coupling guarantee does not survive $|V| \approx 5 \times 10^4$. The strongest calibration and the sharpest limitation share one cause, so an imported guarantee must be checked against the alphabet.

Demoted ECA ordering (§3): the group-mean λ ordering we first computed was never a measurement — five of seven ordered rules return exactly $-0.4 \ln 10$, the constant a log-linear fit yields when damage dies immediately.

Estimator history (§4): the ordinal headline survived a later change to unignited-run handling that moved every mean, which is why it, and not any effect size, is the claim.

NeurIPS Paper Checklist

1. Claims

Question: Do the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract states the contribution as a *validated* black-box instrument and names the three things it does: reproduce known metrics before measuring LMs (the load-bearing rung being Domany–Kinzel, where agreement is *bit-exact* rather than fitted), detect a developmental transition across Pythia checkpoints, and *delimit* itself — real generation absorbs no injected token error, so the quantities describe an iterated-resampling construction rather than decoding. The smooth-limit rungs (logistic map, coupled-map lattice) are labelled in the abstract and body as arithmetic unit tests, not validation. The criticality phenomena are framed as decades-old (measured, not discovered) and cited to prior work. The cross-level proxy hypothesis is reported as a clean *negative*. No capacity scaling law is claimed — that earlier two-seed gap is retracted and explicitly not built upon.

2. Limitations

Question: Does the paper discuss the limitations of the work?

Answer: [Yes]

Justification: The Limitations section states: neither construction samples the model’s actual distribution (masked conditionals are globally inconsistent; the autoregressive rule is consistent but resampled in place on a ring), so all claims are properties of the dynamics; small rings ($N \leq 384$) and one model family for the developmental result, with a second family named as the obvious next requirement; D_{norm} ’s absolute scale doubly qualified (it moves with N , and its numerator’s coupling is not extremal), so only its relative behaviour is quoted; synthetic-only census ground truth; the windowing scheme as a first-class apparatus factor; temperature scales not comparable across constructions. The statistical scope is stated separately, including that λ_{ca} statistics exclude runs whose damage never ignited — for which the quantity is undefined — with n stated in each case.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper contains no formal theorems; it is empirical/measurement work. Connections to the Lieb–Robinson bound and the Bagnoli–Rechtman–Ruffo Lyapunov–velocity relation are cited, not re-derived.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results?

Answer: [Yes]

Justification: All code, model checkpoints (toy), per-run result JSON/NPZ, and figures are supplied as anonymised supplementary material; the Reproducibility appendix gives models, grids, seeds, batch/ring sizes, the diversity-floor and CRN definitions, and the statistical tests. A harness regression test asserts the CRN null coupling is exactly zero.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions?

Answer: [Yes]

Justification: The anonymised supplement contains the full codebase with a README giving per-phase reproduction commands and measured M1 runtime tables, pinned dependency versions, and idempotent resumable scripts. Every number in the paper is traceable to a committed result file; the analysis scripts assert their emitted group sizes against their declared pre-registered design. A public repository URL will be added at camera-ready if the paper is accepted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details?

Answer: [Yes]

Justification: Reproducibility appendix specifies optimizer/steps for the toy model, the exact $(r, T, N, B, \text{sweeps, seeds})$ grids per experiment, the reference corpora, and the metric definitions.

7. Experiment statistical significance

Question: Does the paper report error bars / statistical significance?

Answer: [Yes]

Justification: The paper’s central claims are the validation-ladder reproductions—exact or near-exact matches to known values (the Domany–Kinzel damage identity, which is verified *bit-exactly*—zero mismatching cells—and its critical points to 0.06–0.15%; logistic-map Lyapunov mean error $< 10^{-3}$; census self-TV 0.22 vs cross 0.95)—which are not null-hypothesis tests, so multiple comparisons do not apply to them. Seed error bars are reported where relevant. Where we *do* test—the ECA class separation and the developmental transition—the unit of analysis is the rule or the run (never the seed-cell), the tests are bootstrap or rank-based, and the developmental family is pre-registered and corrected with Benjamini–Hochberg FDR. The finer three-class ECA ordering was tested, failed ($p=0.470$), and is reported as not surviving. The cross-level conclusion is a *negative* reached by sign-consistency across two model families and two within-model axes, not a surviving p -value, so no multiplicity correction is load-bearing. The demoted capacity gap is disclosed as a suggestive two-seed effect and not built upon; an optional seed-level rebuild is tracked in the repository (issues #1–#2, #10).

8. Experiments compute resources

Question: Does the paper provide sufficient information on compute resources?

Answer: [Yes]

Justification: All experiments run on a single Apple M1 Pro (16 GB); per-phase wall-clock runtimes are tabulated. The toy model uses CPU JAX; real models use fp16 on the MPS backend. Total compute is a few laptop-days.

9. Code of ethics

Question: Does the research conform with the NeurIPS Code of Ethics?

Answer: [Yes]

Justification: Measurement/interpretability work on public pretrained models and synthetic data; no human subjects, no sensitive data, no new capability that raises misuse concerns.

10. Broader impacts

Question: Does the paper discuss potential positive and negative societal impacts?

Answer: [Yes]

Justification: See the Responsible use section. Positive: black-box measurement of trained-LM token dynamics without weight access, applicable to closed/API-only models. It creates no new generative capability. The negative case we treat as most likely is misreading: the paper’s damping length and light cone describe an in-place-resampling construction, and real autoregressive generation absorbs no injected token error, so they must not be cited as evidence about deployed-decoder robustness. That boundary is stated in the body, not an appendix.

11. Safeguards

Question: Does the paper describe safeguards for responsible release of high-risk assets?

Answer: [N/A]

Justification: No high-risk models or datasets are released; only measurement code and analysis results on public models.

12. Licenses for existing assets

Question: Are existing assets (code, data, models) properly credited and licenses respected?

Answer: [Yes]

Justification: All assets are publicly released and cited: Pythia and BERT checkpoints (Apache-2.0), GPT-2 (MIT), tinyshakespeare (public-domain source text), WikiText-103 (CC BY-SA 3.0), PyTorch (BSD-3-Clause), HuggingFace Transformers and JAX (Apache-2.0). Usage is research-only and within each licence.

13. New assets

Question: Are new assets introduced in the paper well documented?

Answer: [Yes]

429 Justification: The instrument (code), the synthetic-Markov calibration corpora, and the toy
430 checkpoints are documented in the repository README and the Reproducibility appendix.

431 **14. Crowdsourcing and research with human subjects**

432 Question: N/A.

433 Answer: [N/A]

434 Justification: No human subjects or crowdsourcing.

435 **15. Institutional review board (IRB) approvals**

436 Answer: [N/A]

437 Justification: No human-subjects research.

438 **16. Declaration of LLM usage**

439 Question: Does the paper describe LLM usage if it was an important, original, or non-
440 standard component?

441 Answer: [Yes]

442 Justification: Pretrained LLMs are the object of study — they supply the CA update rule —
443 and this is documented throughout. Separately, and disclosed here per venue policy: an AI
444 coding assistant was used substantially in implementing the experiments, in analysis script-
445 ing, and in drafting. All experimental designs, pre-registrations and claims were reviewed
446 and verified by the authors against committed result files; several errors introduced during
447 that process were caught by the repository’s own assertions and are recorded as retractions
448 in the supplementary findings log.